

In this homework, we will prepare for performing optimal FIR design. Suppose that we are designing a Type I low-pass filter and its order is  $2L + 1$ . In the starter code, you can set up edge frequencies  $\omega_p$ , and  $\omega_s$ . You have two tasks for this homework.

**Task 1 (40%):** For any given  $x_1 < x_2 < \dots < x_{L+2}$ , find a polynomial  $P(x) = a_0 + a_1x + \dots + a_Lx^L$  such that

(i)  $P(x_j) = (-1)^j \delta$  if  $x_j \leq x_s = \cos(\omega_s)$ , and

(ii)  $P(x_j) = 1 + (-1)^j \delta$  if  $x_j \geq x_p = \cos(\omega_p)$ .

Please refer to Eq (7.113) from Oppenheim and Schaffer's book for hints.

Notes: In the starter code, you will find a heuristic way of choosing  $x_1 < x_2 < \dots < x_{L+2}$ . This choice is by no mean optimal, and in the next homework (HW8) we will work on an algorithm that iteratively improves the list to achieve optimality.

**Task 2 (30%):** After finding the polynomial  $P(x) = \sum_{n=0}^L p_n x^n$ , we need to find a zero-phase filter  $h[n]$  such that

$$\begin{aligned} P(\cos \omega) &= H(e^{j\omega n}) = h[0] + 2 \sum_{m=1}^L h[m] \cos m\omega \\ &= h[0] + \sum_{m=1}^L (2h[m]) T_m(\cos \omega) \end{aligned} \tag{1}$$

This can be turned into a problem of solving  $L + 1$  linear equations. Denote the  $m^{\text{th}}$  Chebyshev polynomial as  $T_m(x) = \sum_{n=0}^m w_n^{(m)} x^n$ . Then, Eq. (1) can be formulated as a problem of solving a system of  $L + 1$  linear equations  $Wu = q$ , where

$u = (h[0], 2h[1], 2h[2], \dots, 2h[L])^T$ ,  $q = [p_0, p_1, \dots, p_L]^T$ , and  $W$  is a  $(L + 1) \times (L + 1)$  matrix whose element at the  $n^{\text{th}}$  row and  $m^{\text{th}}$  column is  $w_{n,m} = w_{n-1}^{(m-1)}$  for all  $n, m$ ; in other words, each column represents a Chebyshev polynomial. Your job is to implement a function `myChebyPol()` that returns the matrix  $W$  given any  $L$ . Hint: Use the fact that

$$T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x).$$

**Discussion (30%):** write a report of maximum 2 pages to discuss how the magnitude response may be affected by changing  $\omega_p$ ,  $\omega_s$ ,  $L$  and the way to select  $x_1 < x_2 < \dots < x_{L+2}$ . So far, does this approach perform better than FIR design by windowing? Can you come up with a better way of choosing  $x_1 < x_2 < \dots < x_{L+2}$ ?

Below are a few examples of correct implementation for you to know what to anticipate.

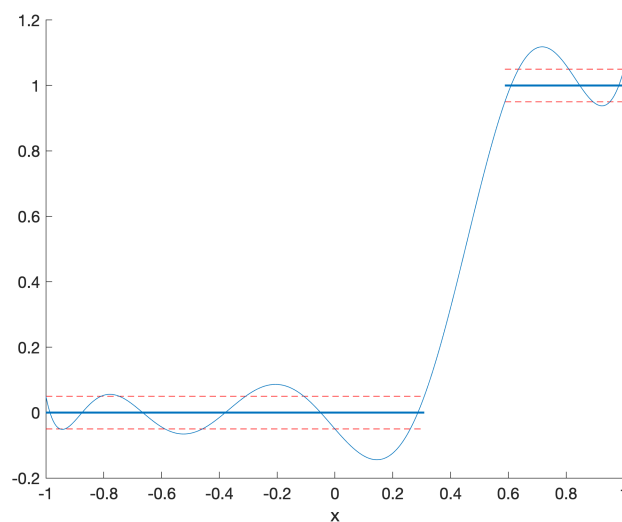


Fig. 1: An example of correct implementation of Task 1.

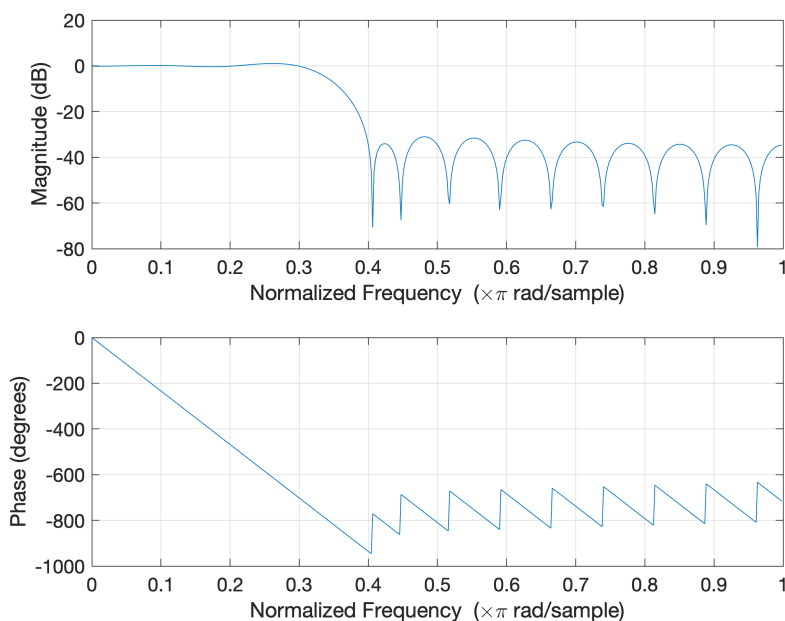


Fig. 2: An example what to expect when Task 2 is also correctly implemented.